

controlled rough path norm of the solution, [GVR25, BNGVS25] imposed that $\sigma \in [0, \frac{1-\gamma}{2})$. In this work we are able to overcome this limitation and treat the case $\sigma \in [0, \gamma)$ as well.

We mention that the techniques developed in this work are also applicable to nonlinear diffusion coefficients G that are further assumed to be $N + 1$ -times Fréchet differentiable with bounded derivatives. Here $N = \lfloor \frac{1}{\gamma} \rfloor$ represents the number of iterated integrals that have to be taken into account. For a better comprehension we only focus here on the linear case and work with the mild formulation of (1.1). For well-posedness results of semilinear rough stochastic partial differential equations with linear diffusion coefficients based on a variational approach we refer to [FNS20].

This manuscript is structured as follows. In Section 2 we introduce the main concepts of controlled rough paths and control functions required in this setting. We specify that we consider (p, γ) -rough paths $\mathbf{X}$ for $p < \gamma$ and define suitable controlled rough paths according to $\mathbf{X}$. Our definition is a natural generalization of the controlled rough paths according to a monotone scale of interpolation spaces introduced in [GHN21]. Here we incorporate in Definition 2.6 more iterated integrals and decrease the spatial regularity of the Gubinelli derivatives and remainders according to the time regularity γ of the rough path $\mathbf{X}$. The main innovative aspect in comparison to [GHN21] is the replacement of the corresponding Hölder norms of the rough input $\mathbf{X}$ and of the controlled rough path with suitable control functions. These are necessary in order to establish the desired integrable bounds and represent a main technical ingredient of our work. To our best knowledge, this is the first work that introduces controlled rough paths measuring their time regularity in suitable Banach spaces in terms of control functions instead of Hölder continuity. Afterwards we construct a rough integral for such controlled rough paths against $\mathbf{X}$. This is naturally achieved by means of a modified version of the Sewing lemma stated in Theorem 2.11, keeping track of several parameters reflecting an interplay between the time regularity of $\mathbf{X}$, the loss of spatial regularity $\sigma < \gamma$, the number of iterated integrals and the parameter $p < \gamma$ arising in the definition of the (p, γ) -rough path chosen such that $\sigma < p < \gamma$ and of the control function. As already mentioned, the main technical challenge is to incorporate the control functions in our arguments.

Section 3 establishes the well-posedness of (1.1) based on the properties of the rough integral introduced in Section 2. This can be achieved by a standard fixed-point argument in the space of controlled rough paths introduced in Section 2. We provide the necessary steps required in order to set up this argument relying on the assumption that the diffusion coefficient G is linear. As already mentioned, one could incorporate a nonlinear term G which is $N + 1$ Fréchet differentiable with bounded derivatives. In this case, our techniques combined with a standard fixed-point argument as in [GH19, GHN21] entail the local well-posedness of (1.1). An additional boundedness assumption on the Fréchet derivatives up to order $N + 1$ as in [HN22] will imply the global-in-time existence. Such boundedness assumptions have been relaxed for rough differential equations in [LY25]. We refrain from providing the details of the fixed-point argument and focus instead on establishing moment bounds for (1.1).

Section 4 is inspired by the seminal paper [CLL13] that establishes the existence of moments of all order for the norm of differential equations driven by Gaussian rough paths. We extend this result in Theorem 4.17 to the infinite-dimensional case based on Borell's inequality and a detailed analysis of the iterated integrals arising in our setting. Moreover we provide suitable estimates for the translated path in terms of the control function introduced in Section 2. We emphasize that